

NBC IDEA FACTORY SEASON 10

COLLEGE- NATIONAL INSTITUTE OF TECHNOLOGY JAMSHEDPUR

TEAM- MONTY BHAI

SALES AFTERMARKET INTELLIGENCE PLATFORM

Predictive Sales through Connected Customer & Asset Data

The objective of this project is to discuss a Smart Aftermarket Intelligence Platform that leverages connected customer and product data to predict aftermarket demand using a multi-score, probability-based approach for proactive sales and service decisions for bearings, adjacencies, and services is discussed further.

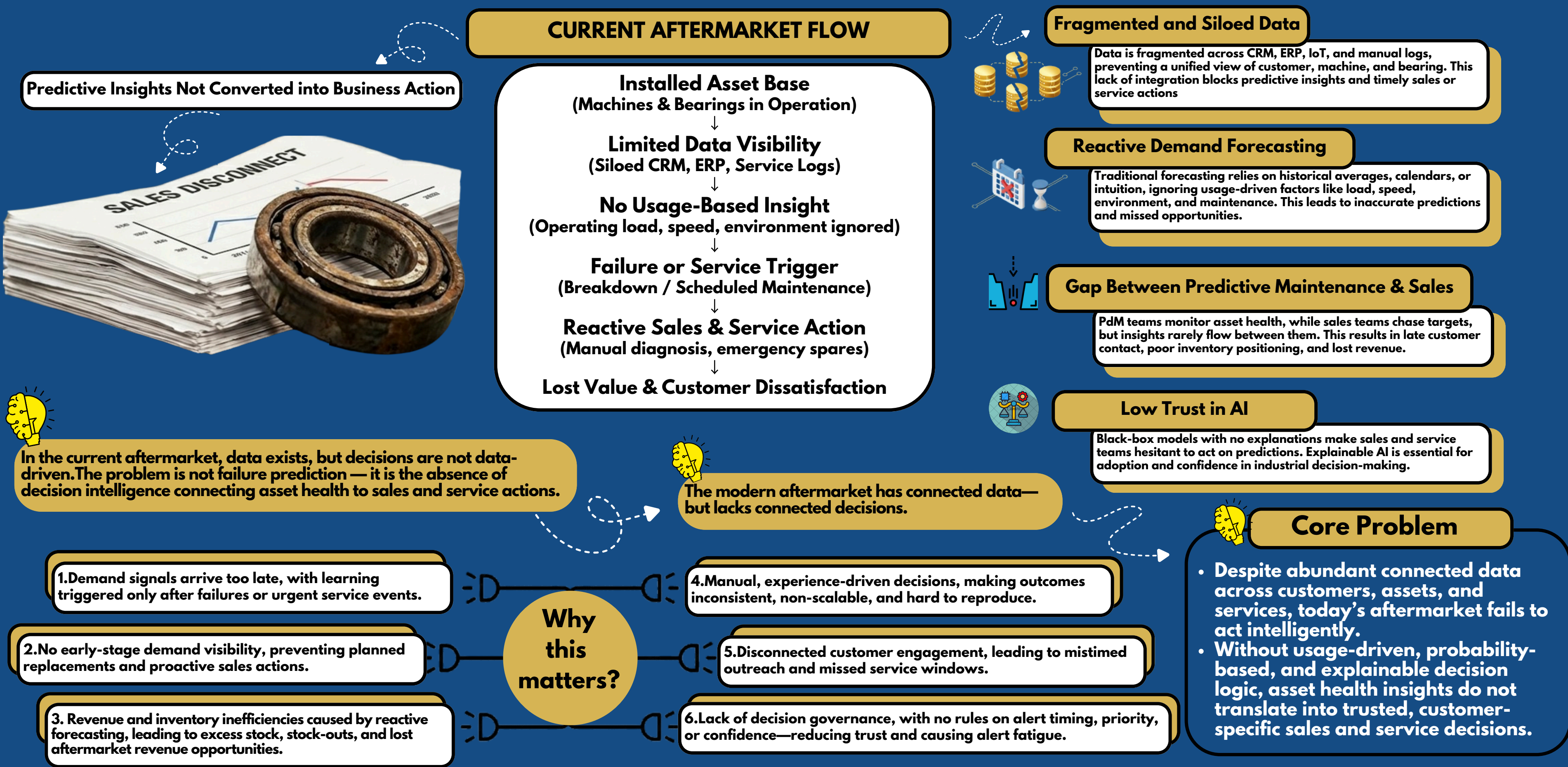
EXPECTED BUSINESS VALUE

- 12-18% increase in aftermarket revenue driven by predictive, probability-based sales targeting.
- 15-30% improvement in sales and service productivity through smart notifications and opportunity prioritization.
- 10-20% reduction in inventory costs and unplanned downtime using connected customer and asset insights.

TEAM MEMBERS

REHAAN RAZA	2023UGMM052
ANKIT KUMAR	2023UGMM080
ATUL SHAKTI	2023UGMM036

WHY TRADITIONAL AFTERMARKET FAILS?



INNOVATION AND ITS FEASIBILITY

WHY AIM³?

Traditional Systems	AIM ³ Platform
Predict failures	Predict demand
Asset-centric	Customer-centric
Black-box AI	Explainable decisions
Dashboards	Action & notifications
Reactive sales	Predictive sales

BUSINESS AND ECONOMIC FEASIBILITY

- No new hardware dependency
- Leverages existing CRM, ERP (SAP), IoT, service logs
- Software-first investment
- Low CAPEX, scalable OPEX model
- Uses already-collected data
- Monetizes sunk data costs
- Modular deployment
- Can start with top customers / critical assets

AIM³ — Aftermarket Intelligence & Monetization Engine

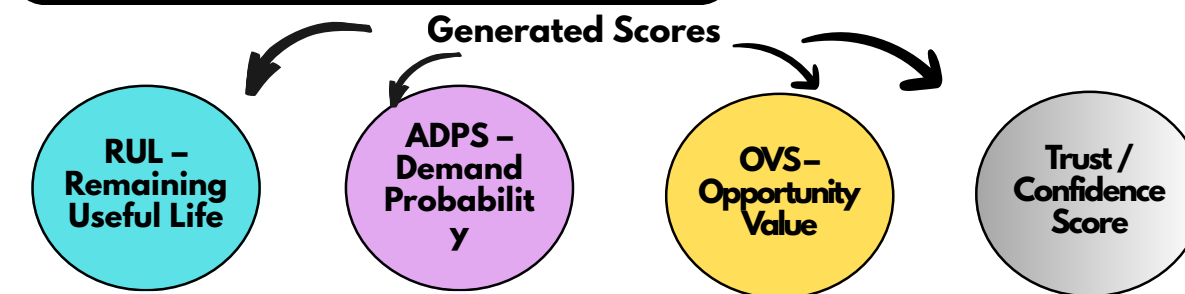
LAYER 1: DIGITAL THREAD

- Connected Lifecycle Intelligence
Customer → Machine → Bearing → Sensor → Health → Demand
- Contextualized data, not raw signals

LAYER 2 — PREDICTIVE INTELLIGENCE ENGINE

- Predictive Sales Intelligence Engine
 - Predicts probability of demand, not just failure
 - Time windows: 7 / 30 / 90 days
 - Severity & urgency assessment
 - Confidence-aware predictions
- Hybrid Modeling
Physics-based rules + ML (LSTM / RF/Regression)

LAYER 3 — DECISION SCORES ENGINE



LAYER 4 — INTELLIGENT NOTIFICATION & ENGAGEMENT SYSTEM

- Context-Aware Notification Engine
Probability-Based Engagement
- LOW MEDIUM HIGH
- Monitor only (no notification) : AI-generated soft reminder : Sales + Service outreach

LAYER 5 — ACTION & MONETIZATION

- Triggered Actions
 - CRM sales lead creation
 - Service scheduling
 - Inventory repositioning
 - Dynamic pricing / bundling
- Revenue Models
 - Predictive service subscriptions
 - Usage-based contracts
 - Outcome-based pricing

CUSTOMER RESPONSE & SERVICE OUTCOME FEEDBACK

QUANTIFIED BUSINESS BENIFITS

- Aftermarket Revenue Uplift: +12–18%
 - Driven by ADPS + OVS-based prioritization
- Unplanned Downtime Reduction: 20–35%
 - Predictive 7 / 30 / 90-day interventions cut surprise failures, improving asset availability (+3–5%) and throughput (+2–4%)
- Inventory Optimization: 10–25% Working Capital Reduction
 - Probability-based stocking cuts slow inventory (10–15%) and boosts inventory turns (15–25%).
- Customer Retention Improvement: +5–10%
 - Transparent, advisory notifications improve retention by 5–10%, increase repeat orders by 10–15%, and raise customer lifetime value (CLV).

WHAT'S THE PREDICTION AND NOTIFICATION

- Predictiveness of AIM³
 - WHAT → Bearing, Adjacencies Service/replacement need
 - WHEN → Next 7 / 30 / 90 days
 - WHY → Load, vibration trend, usage severity, service history

From prediction → probability → notification → revenue
 AIM³ does not predict failures — it predicts revenue opportunities with confidence.

